
Vraic to the Future

VRAIC FUTURES ENGINE

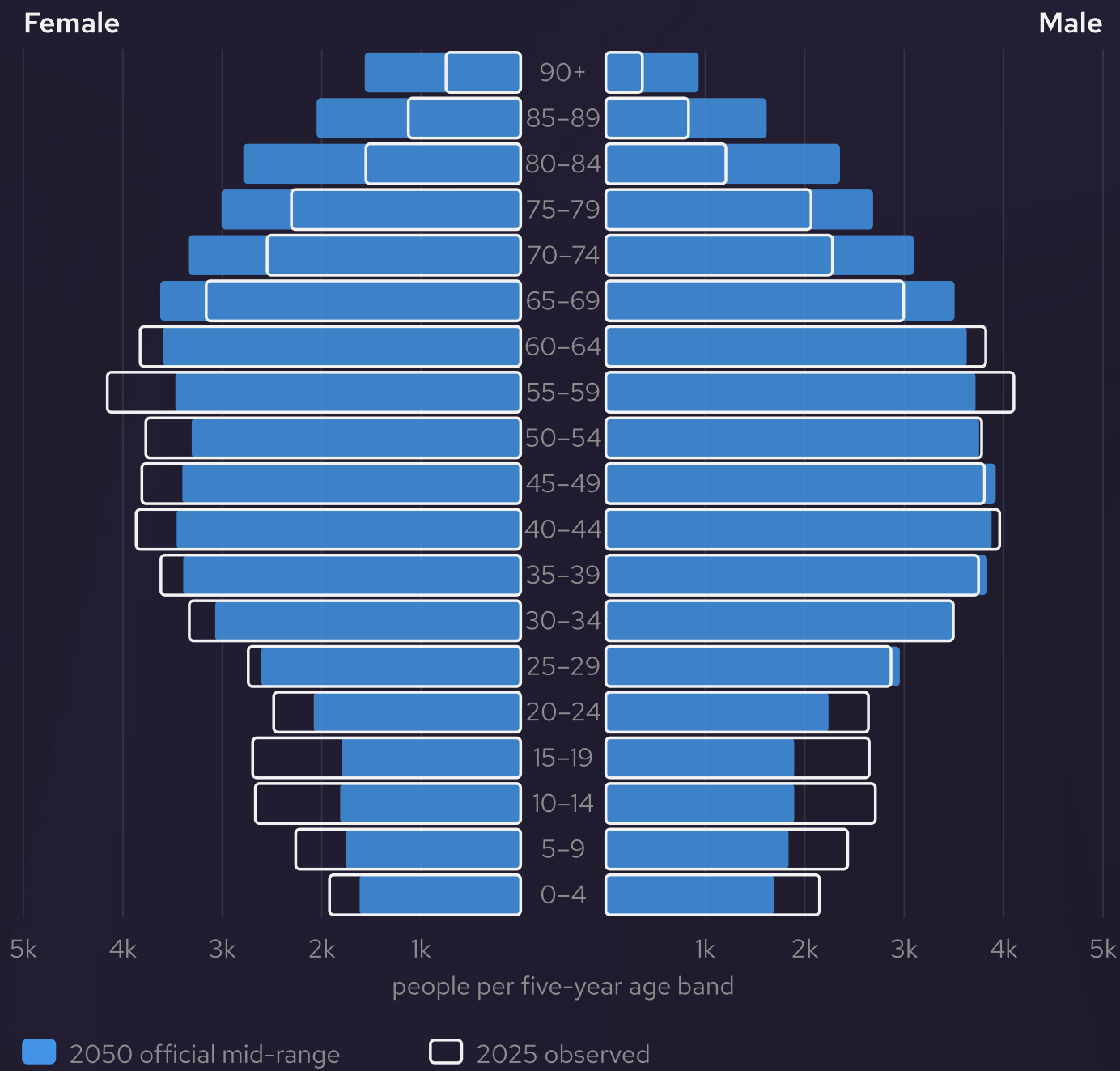
Conversational, probabilistic population scenarios for Jersey



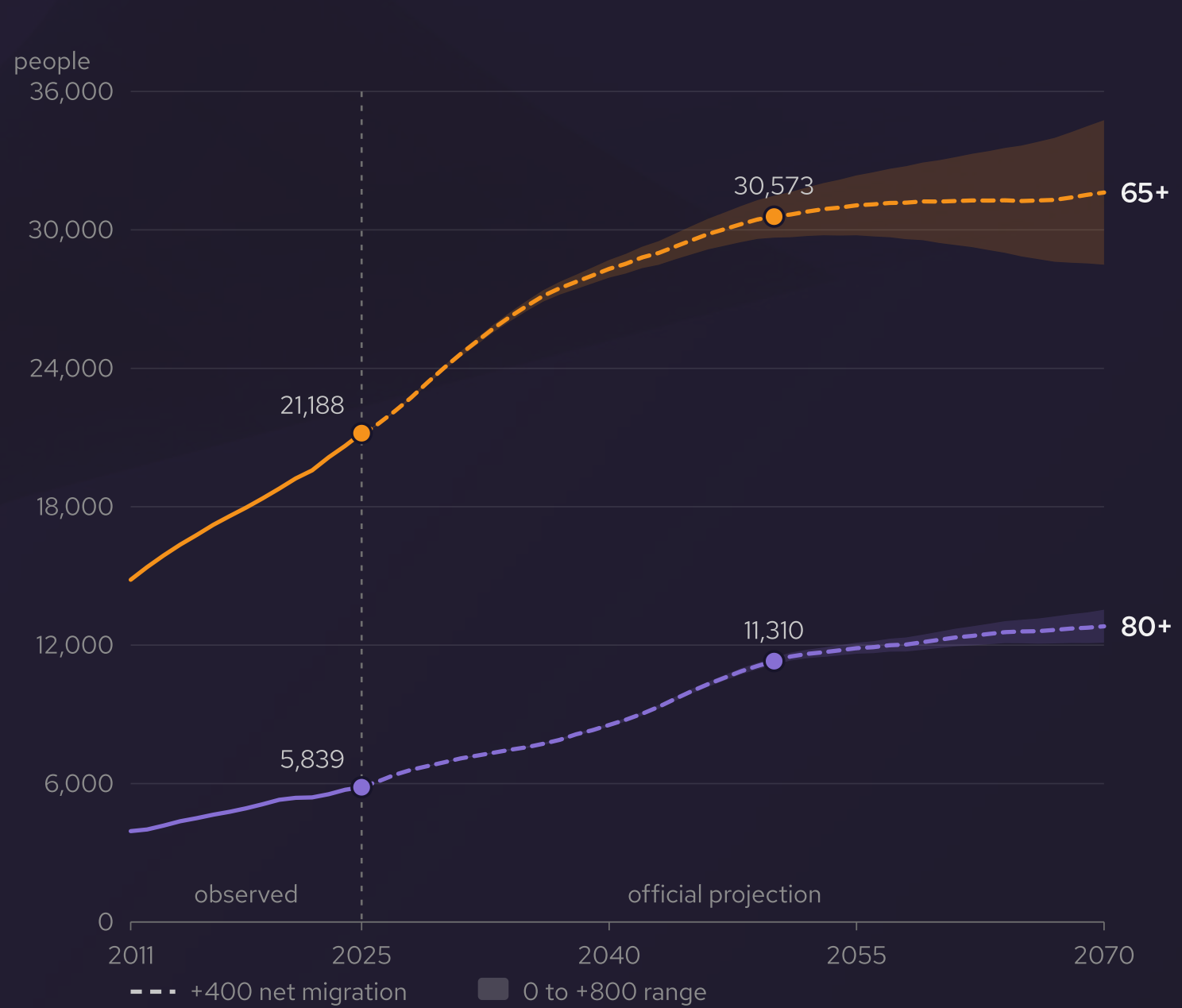
Jersey is ageing

5,839 to 11,310
residents aged 80+, 2025 to 2050

Population by age and sex, 2025 and 2050



Residents aged 65+ and 80+



Ageing strains public finances and the workforce

JERSEY

≈ +700

net migration a year to keep the 16–64 population stable to 2080, at mid-range fertility¹

2029

Long-Term Care Fund exhausted on current plans²

2072

Social Security Fund exhausted under Budget 2026 (was 2082); about 10 years sooner with lower migration³

INTERNATIONAL

8% to 13%

UK health spending, % of GDP, 2030–31 to 2075–76, OBR baseline⁴

–12%

EU labour force, 2022 to 2070: 25 million people⁵

3.0 to 1.9

people aged 20–64 per person aged 65+ across the OECD, 2025 to 2050⁶

1. Statistics Jersey (2026). Jersey population projections 2025 to 2080, §2 and §5.2

2. Fiscal Policy Panel (2025). Annual Report 2025, p. 5

3. Government Actuary's Department (2026). Jersey Social Security Fund budget proposals impact

4. Office for Budget Responsibility (2026). Fiscal risks and sustainability, July 2026, p. 5

5. European Commission (2024). 2024 Ageing Report, Institutional Paper 279, p. 4

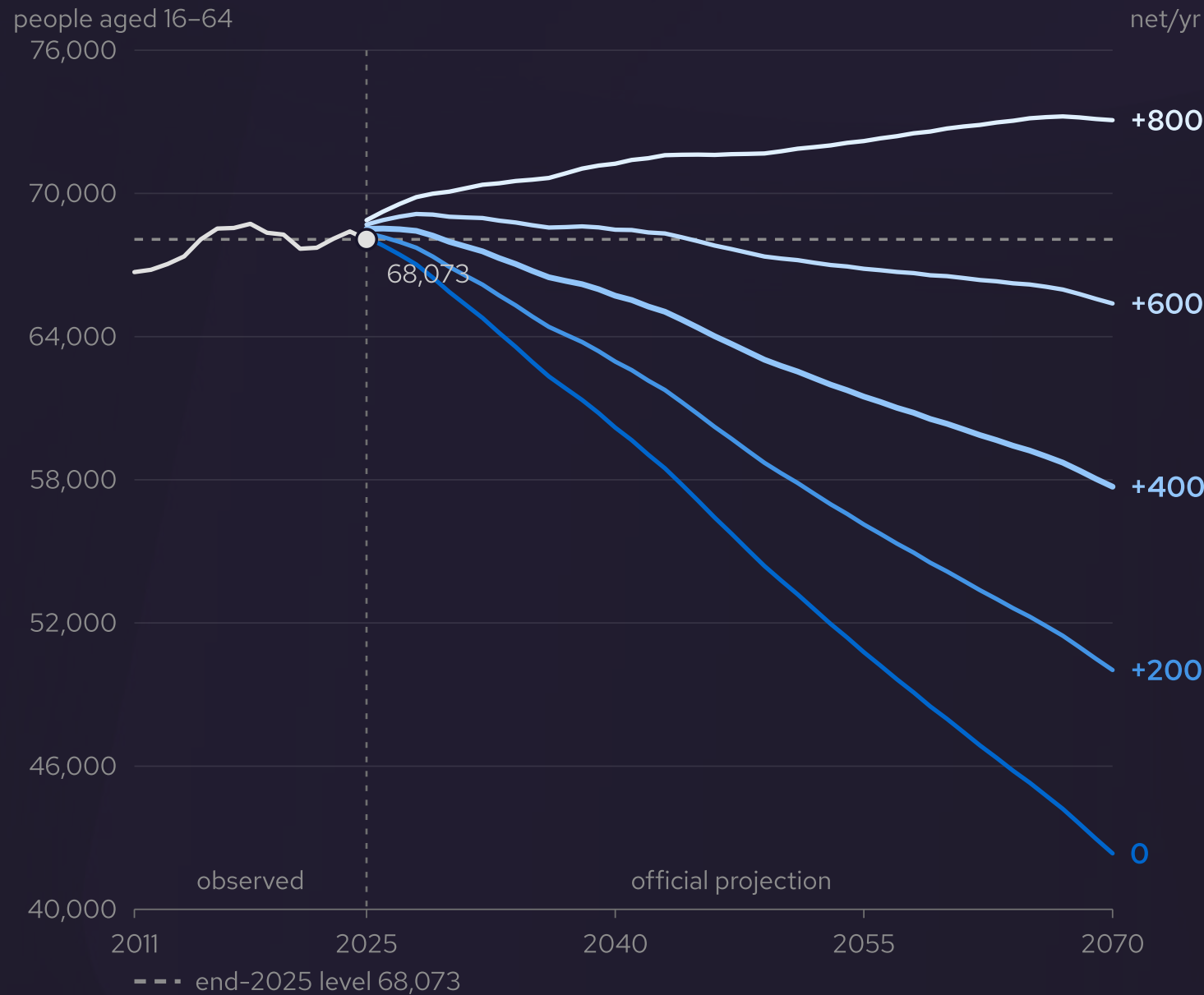
6. OECD (2025). Pensions at a Glance 2025, chapter 1, p. 14; reciprocal of 33 and 52 aged 65+ per 100 aged 20–64

Fewer people of working age for every older resident

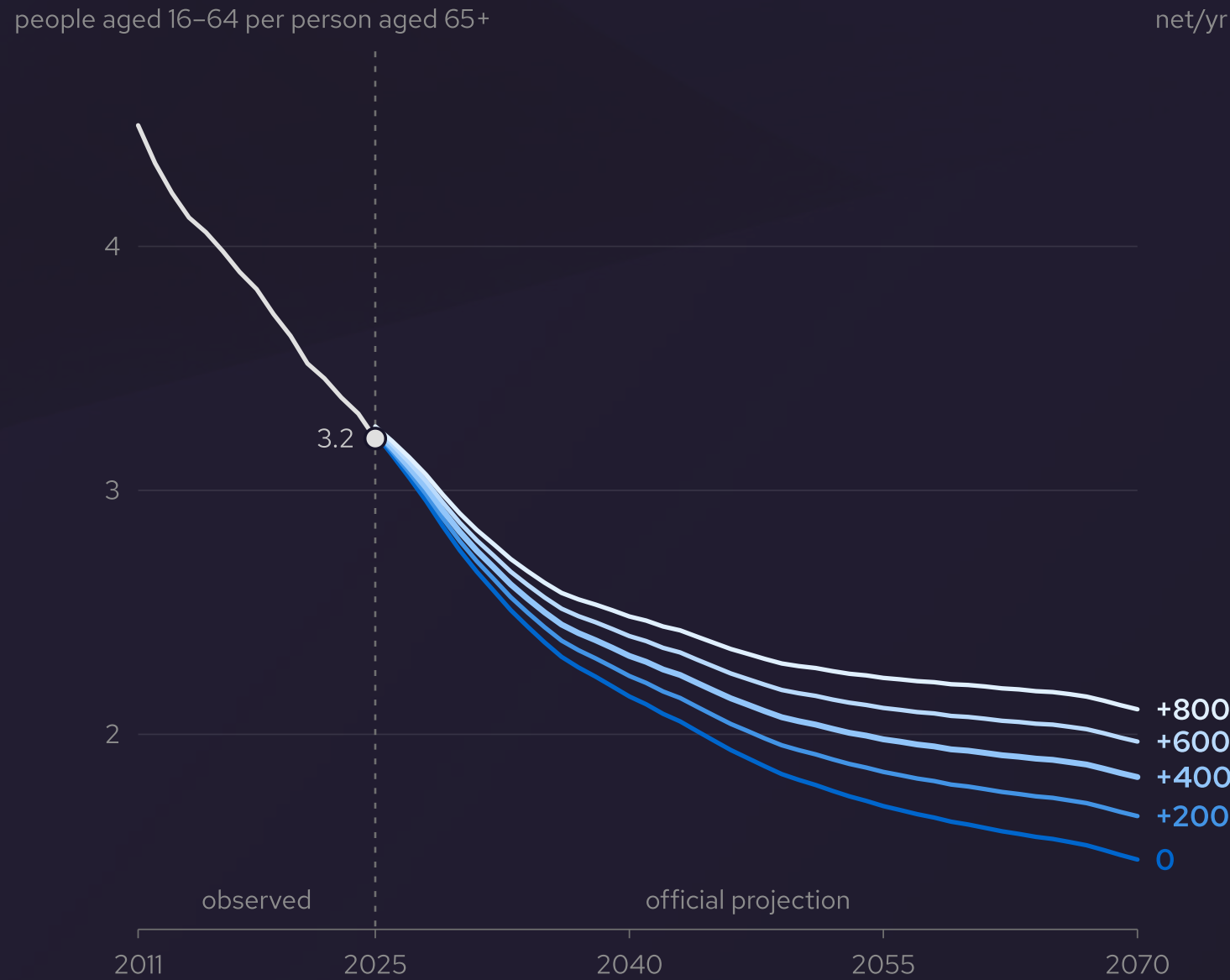
3.2 to 2.1

people aged 16–64 per person aged 65+, 2025 to 2050, +400 a year

Residents aged 16–64, by annual net migration



People aged 16–64 per person 65+, by net migration



Ageing reshapes every public service

WORKFORCE

2.1

people aged 16–64 per
person aged 65+ in 2050
3.2 in 2025

16–64	68,073 to 62,750
65+	21,188 to 30,380

HEALTH AND CARE

+26%

hospital bed days by 2040
62,560 to 78,620

65+ care claims, 2040	+43%
GP appointments, 2040	+7%

EDUCATION

–21%

primary pupils by 2035
7,113 to 5,594

Secondary, 2040	–26%
Aged 0–15, 2040	–24%

HOUSING

2,680

extra homes needed by
2040

90% range	210–5,190
Population, 2040	+1%

How it works

Ask

local LLM drafts the change

Review

validated assumptions, you confirm

Simulate

10,000 futures: births, deaths,
migration, rates

Answer

median, 90% range, chance

"Keep everything else the same, but make migration zero after 2030."

62,100

aged 16–64 in 2040
90% range 57,800–66,500

–3,500

against the +400 a year baseline

99%

chance of fewer people of working
age than at end-2025 (68,073)

Same cohort-component method as Statistics Jersey

Reproduces their projections

10,000 simulations per question

Ranges checked against what happened

What if? What would it take?

WHAT IF · OFFICIAL +400 A YEAR

"What is the chance that the working-age population is below today's level in 2040?"

83%

fixed assumptions: 100%

WHAT WOULD IT TAKE

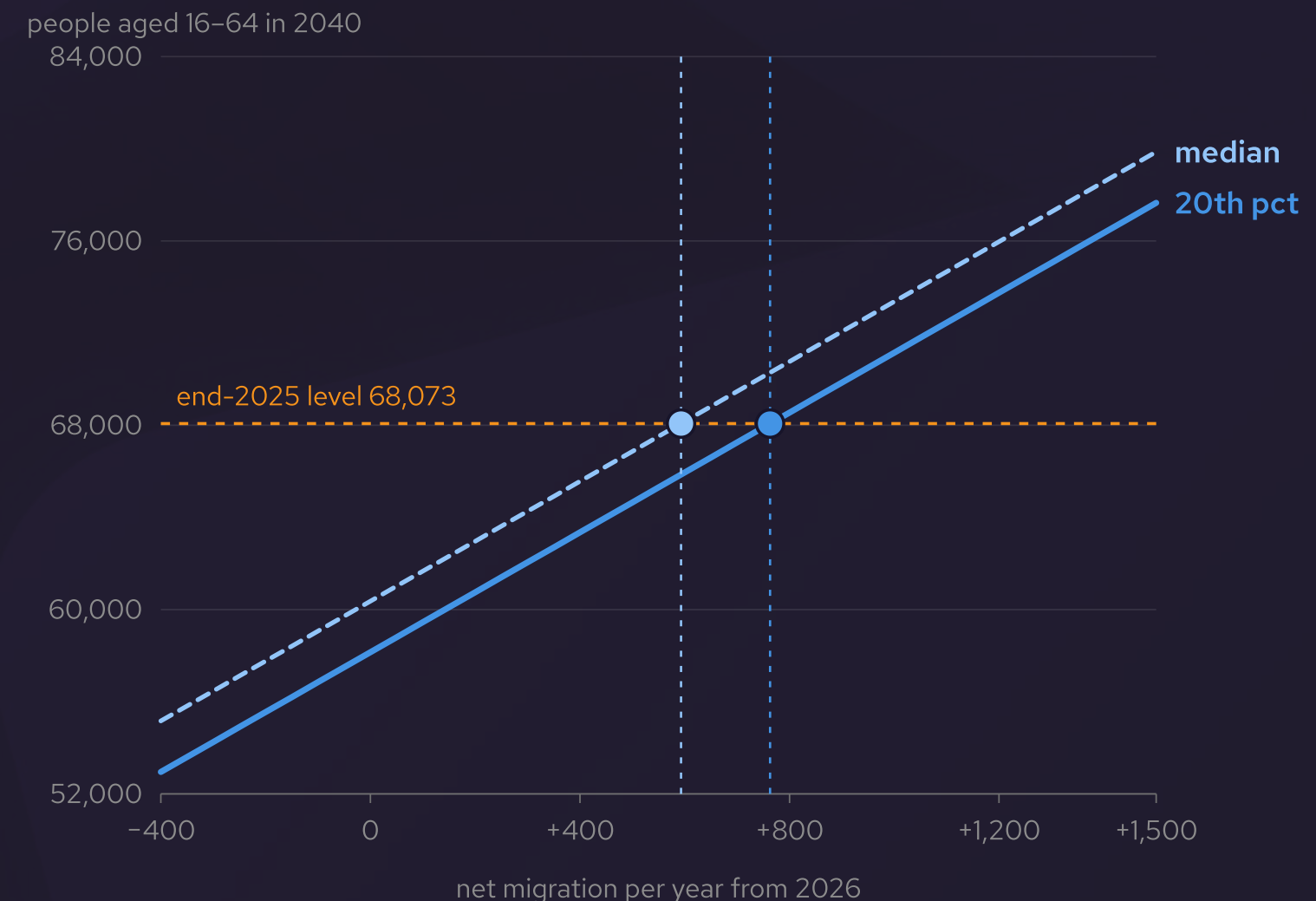
"What net migration keeps the number of people of working age per person aged 65+ at today's level by 2040?"

Not reachable

2.7 at +1,500; today 3.2

WHAT WOULD IT TAKE

"What net migration gives an 80% chance of keeping the working-age population at today's level by 2040?"



+760 a year; the median alone needs +593

Economy, health, schools, housing

ECONOMY AND SKILLS

"What is the chance of fewer than two people of working age per person aged 65+ by 2050?"

1 in 3 31% of 10,000 simulations; median 2.1 per person aged 65+; 90% range 1.9–2.3

HEALTH AND CARE

"How many nurses and midwives would 2040 need at today's staffing per bed day?"

1,397 FTE; 90% range 1,315–1,479; end-2025: 1,112

EDUCATION

"How many secondary pupils should we plan for in 2040?"

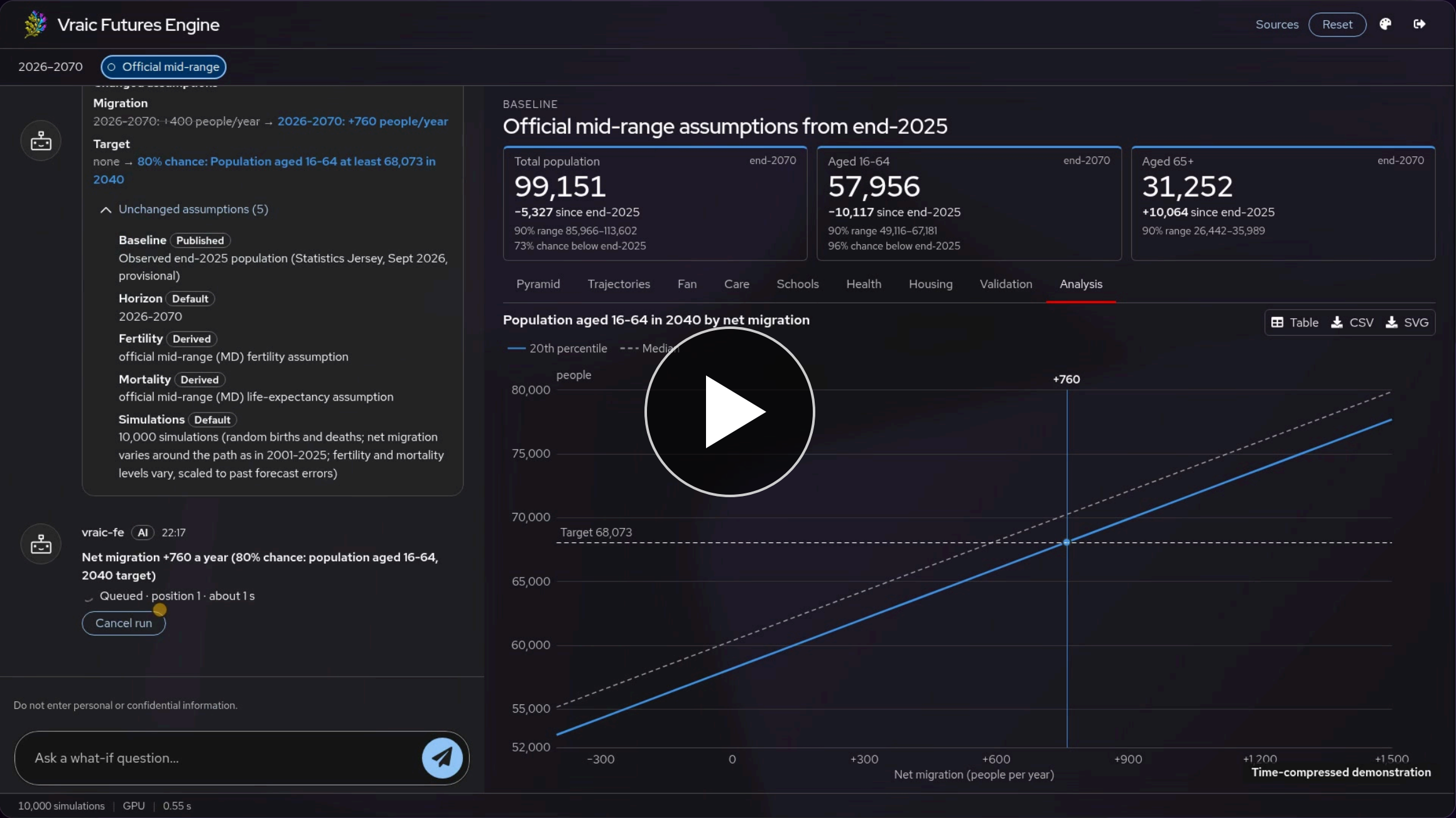
4,817 90% range 4,519–5,124; end-2025: 6,474

HOUSING

"What net migration is consistent with 2,000 additional homes by 2040?"

+314 a year

Demonstration



vraic.nat.je

ACCESS CODE

quartz-raven-982

A six-week pilot

WEEKS 1-2

Agree the baseline and methods with Statistics Jersey: statistical review

WEEKS 3-4

Trial with users from education, health, housing and other departments

WEEKS 5-6

Connect the Government's Copilot through an MCP server: every employee sees how demographics affect their department over time

ONGOING

Add datasets as users find what the modelling in their domain needs



VRAIC FUTURES ENGINE

github.com/gnathoi/vraic-fe